

EXPERT RF DESIGN: TRIANGULAR GRID ARCHITECTURE

1. THE TRIANGULAR CELLULAR CONCEPT

The Triangular Grid placement involves positioning Access Points (APs) at the vertices of equilateral triangles. This creates a 'Honeycomb' coverage pattern.

Unlike square layouts, the distance between any point in the triangle and the nearest three APs is minimized, ensuring that there are no 'weak signal' valleys in the center of the grid.

2. RATIONALE FOR VOICE & FAST MOBILIZATION

- VOICEOVER WI-FI (VoWiFi): Voice packets are jitter-sensitive. A triangular layout ensures a constant Signal-to-Noise Ratio (SNR) of $>25\text{dB}$, which is mandatory for clear, jitter-free audio.

- FAST ROAMING (MOBILIZATION): By maintaining a 20% overlap between APs in a triangle, mobile clients can perform pre-authentication (802.11i/r) seamlessly. The client's 'roaming decision' is faster because the secondary and tertiary signals are always available at predictable signal strengths.

3. EXPERT DESIGN PARAMETERS (LLD STANDARDS)

- RSSI Threshold: -65 dBm (Primary Coverage)
- Secondary RSSI: -67 dBm (For seamless 802.11r handover)
- Channel Width: 20MHz (Avoid 40/80MHz in high-density to prevent Co-Channel Interference)
- Minimum SNR: 25 dB
- Packet Loss: $<1\%$
- Jitter: $<10\text{ms}$

4. AP PLACEMENT & FREQUENCY REUSE

APs should be staggered. For example:

Row 1: AP1 (Ch 1), AP2 (Ch 6), AP3 (Ch 11)

Row 2: (Offset by 50%) AP4 (Ch 11), AP5 (Ch 1), AP6 (Ch 6)

This prevents 'Adjacent Channel Interference' and ensures that every vertex of the triangle has a different non-overlapping channel.

5. INDUSTRY REFERENCES & STANDARDS

- IEEE 802.11r-2008 (Fast Basic Service Set Transition)
- IEEE 802.11k-2008 (Radio Resource Management)
- Cisco Wireless High-Density Design Guide (CVD)
- Ekahau RF Planning Best Practices for Voice & Roaming